



Sulfidated nano zerovalent iron (S-nZVI) for *in situ* treatment of chlorinated solvents: A field study

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ABSTRACT

Sulfidated nano zerovalent iron (S-nZVI), stabilized with carboxymethyl cellulose (CMC), was successfully synthesized on site and injected into the subsurface at a site contaminated with a broad range of chlorinated volatile organic compounds (cVOCs). Transport of CMC-S-nZVI to the monitoring wells, both downgradient and upgradient, resulted in a significant decrease in concentrations of aqueous-phase cVOCs. Short-term (0–17 days) total boron and chloride measurements indicated dilution and displacement in these wells. Importantly however, compound specific isotope analysis (CSIA), changes in concentrations of intermediates, and increase in ethene concentrations confirmed dechlorination of cVOCs. Dissolution from the DNAPL pool into the aqueous phase at the deepest levels (4.0–4.5 m bgs) was identifiable from the increased cVOCs concentrations during long-term monitoring. However, at the uppermost levels (~1.5 m above the source zone) a contrasting trend was observed indicating successful dechlorination. Changes in cVOCs concentrations and CSIA data suggest both sequential hydrogenolysis as well as reductive β -elimination as the possible transformation mechanisms during the short-term abiotic and long-term biotic dechlorination. One of the most positive outcomes of this CMC-S-nZVI field treatment is the non-accumulation of lower chlorinated VOCs, particularly vinyl chloride. Post-treatment soil cores also revealed significant decreases in cVOCs concentrations throughout the targeted treatment zones. Results from this field study show that sulfidation is a suitable amendment for developing more efficient nZVI-based *in situ* remediation technologies.

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1. Introduction

Sulfidation is a recent development related to the use of zerovalent iron (ZVI) based materials for groundwater remediation

(Fan et al., 2017; Li et al., 2017). Though most of the focus in recent years has been on engineered sulfidation of nano ZVI (nZVI), biogenic sulfidation of ZVI has been extensively investigated since the 1990s (Benner et al., 2002; Phillips et al., 2000; Puls et al., 1999; Wilkin et al., 2003). Earlier studies focused on the identification of authigenic mineral phases formed during the application of ZVI permeable reactive barriers (PRBs). Formation of iron sulfides (FeSs) was attributed to the concurrent oxidation of ZVI and generation of sulfides via biogeochemical processes (e.g., microbial reduction of SO_4^{2-}). These mineral phases were studied in the

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